

Smart Parking Sensor

Oğuzhan Bursa
Turkish-German University (TAÜ)

May 23, 2026

1 Introduction

This project is a bare-metal C implementation of a smart parking sensor system using the MSP430F5529 microcontroller. The goal of the project is to measure the distance of an object using an ultrasonic sensor and provide both visual and auditory feedback based on the proximity of the object.

2 Hardware Requirements

The following components were used in the development of this project:

- MSP430F5529 LaunchPad
- HC-SR04 Ultrasonic Distance Sensor
- Passive Buzzer
- 3x LEDs (Red, Yellow, Green)
- 3x 220 Ω Resistors (for LEDs)
- 2x 10 k Ω Resistors (for logic level shifting on the Echo pin)
- Breadboard and Jumper Wires

3 Implementation Details

The system relies on hardware timers rather than software delays to ensure accurate distance measurement and non-blocking sound generation.

3.1 Distance Measurement

The HC-SR04 sensor is triggered by sending a 10 μ s HIGH pulse from pin P1.6. The sensor responds with a HIGH pulse on the Echo pin, the duration of which is proportional to the distance of the object. A voltage divider circuit (using two 10 k Ω resistors) is used to safely step down the 5V Echo signal to 2.5V for the MSP430's 3.3V logic. The Timer_A1 module in Input Capture mode (P2.0) is utilized to measure the exact pulse width via hardware interrupts.

3.2 Feedback Mechanism

- **Safe Zone (> 30 cm):** LEDs and buzzer are turned off.
- **Far (20 – 30 cm):** Green LED is on, buzzer beeps slowly.
- **Middle (10 – 20 cm):** Yellow LED is on, buzzer beeps faster.
- **Close (< 10 cm):** Red LED is on, buzzer emits a continuous sound.

The auditory feedback is generated using the Timer_A0 module to output a PWM signal to the passive buzzer on pin P1.2, adjusting the duty cycle to turn the sound on and off without blocking the main CPU loop.

4 Conclusion

This project successfully demonstrates the use of hardware timers, interrupts, and PWM generation on the MSP430F5529 using bare-metal C programming. The system operates efficiently and provides accurate real-time feedback for distance measurement.